

MONT JOLI, QC, Canada

WMO: 717180

Lat: 48.603N

Lon: 68.210W

Elev: 172

StdP: 14.60

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.4	(c) -5.1	(d) -19.0	(e) 2.0	(f) -8.9	(g) -14.7	(h) 2.5	(i) -4.3	(j) 35.9	(k) 17.7	(l) 31.2	(m) 14.9	(n) 13.7	(o) 270	(p) 0.704

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	80.6	68.0	77.4	66.0	74.7	64.4	69.7	77.6	67.7	75.3	65.9	72.9	12.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	98.4	74.6	64.6	91.9	72.6	62.8	86.1	70.6	33.9	77.6	32.3	75.4	30.8	72.9	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.8	25.4	22.8	-14.7	87.6	5.4	3.0	-18.5	89.8	-21.7	91.6	-24.7	93.3	-28.6	95.4
			-15.2	73.5	5.3	2.4	-19.0	75.2	-22.1	76.6	-25.1	77.9	-28.9	79.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.6	12.6	14.6	23.7	36.0	48.3	58.5	64.6	63.1	55.2	43.7	32.8	20.9
	DBStd	19.68	11.17	10.20	9.44	7.05	6.98	7.02	5.52	5.47	6.93	7.02	8.19	9.30
	HDD50	5182	1160	992	816	422	118	8	0	0	24	221	518	903
	HDD65	9401	1625	1412	1280	869	521	220	76	102	304	659	965	1368
	CDD50	1394	0	0	0	2	64	139	451	406	179	27	3	0
	CDD65	139	0	0	0	0	2	23	62	43	9	0	0	0
	CDH74	741	0	0	0	0	24	148	309	211	48	1	0	0
	CDH80	110	0	0	0	0	3	28	43	30	6	0	0	0
	WSAvg	10.8	12.8	12.4	12.1	11.3	10.0	9.1	8.7	8.7	9.4	10.8	12.0	12.5
	PrecAvg	38.1	3.5	2.8	2.8	2.3	3.2	2.8	3.1	3.5	3.3	3.2	3.3	4.0
Precipitation	PrecMax	51.2	8.3	8.1	6.1	5.4	6.0	6.6	6.9	8.1	6.1	5.3	7.5	6.0
	PrecMin	29.8	1.4	0.4	0.9	0.5	0.7	0.6	1.5	1.2	1.0	1.2	1.6	1.7
	PrecStd	4.6	1.5	1.4	1.5	1.2	1.3	1.4	1.1	1.6	1.4	1.2	1.3	1.3
	PrecStd	4.6	1.5	1.4	1.5	1.2	1.3	1.4	1.1	1.6	1.4	1.2	1.3	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.4	44.3	50.7	61.6	77.3	83.3	84.3	83.6	79.1	69.6	62.9	45.8
		MCWB	42.5	41.8	45.4	50.7	61.0	67.3	70.2	69.0	67.8	60.1	57.6	42.4
	2%	DB	37.9	36.1	42.9	54.4	70.5	78.0	80.5	79.2	74.1	63.9	53.5	40.0
		MCWB	35.7	33.5	38.0	45.2	56.9	64.8	68.3	67.5	64.8	56.4	49.0	37.6
	5%	DB	33.2	32.7	39.4	49.4	64.6	74.1	77.4	75.9	70.1	58.7	48.0	36.4
		MCWB	31.5	31.2	35.4	41.9	54.3	62.9	66.4	65.7	62.4	52.7	43.5	34.2
	10%	DB	28.7	29.8	36.1	45.6	60.1	70.7	74.6	72.9	66.6	54.5	43.5	33.5
		MCWB	27.0	28.2	32.7	39.9	51.2	60.9	64.8	63.9	59.9	49.4	40.1	31.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.9	41.6	47.2	52.8	64.0	70.6	73.1	71.4	69.9	61.6	57.7	43.2
		MCDB	44.8	44.0	50.1	59.7	74.4	80.4	80.7	81.0	76.9	67.7	61.9	45.1
	2%	WB	35.9	34.2	38.7	46.3	59.3	67.1	70.1	68.8	66.1	57.1	49.6	37.9
		MCDB	37.9	35.4	41.8	52.2	67.1	75.1	77.8	77.0	72.2	63.1	53.1	39.5
	5%	WB	31.5	31.4	35.7	43.1	55.4	64.4	68.1	67.1	63.3	53.4	44.1	34.5
		MCDB	33.2	32.6	38.9	48.4	63.9	71.5	75.4	74.2	69.4	58.2	47.0	36.3
	10%	WB	27.1	28.4	33.1	40.2	52.0	62.0	66.1	65.2	60.3	49.9	40.5	32.0
		MCDB	28.7	29.7	35.6	44.6	58.8	69.1	73.2	71.5	66.0	53.8	43.1	33.2
Mean Daily Temperature Range	5% DB	MDBR	13.2	13.1	12.8	12.8	16.0	16.7	15.9	15.7	14.7	11.5	9.9	10.6
		MCDBR	18.3	14.2	15.8	19.5	23.4	21.5	19.0	19.2	19.5	18.2	16.4	14.0
		MCWBR	17.6	13.0	12.8	13.4	13.7	12.6	10.7	11.1	12.8	13.1	14.0	13.4
	5% WB	MCDBR	18.1	13.4	14.2	18.4	21.7	19.2	17.8	17.2	19.0	17.5	15.7	13.8
		MCWBR	17.5	12.6	12.2	13.6	13.9	12.7	11.7	11.2	13.7	13.5	14.2	13.4
Clear-Sky Solar Irradiance	taub		0.261	0.269	0.300	0.359	0.367	0.388	0.414	0.393	0.367	0.329	0.298	0.271
	taud		2.341	2.312	2.286	2.287	2.352	2.352	2.314	2.374	2.426	2.510	2.460	2.364
	Ebn at Noon		259	284	291	282	284	277	267	268	265	259	243	238
	Edh at Noon		22	28	34	38	37	38	39	35	30	23	20	19
All-Sky Solar Radiation	RadAvg		395	688	1059	1395	1632	1742	1704	1461	1118	656	351	264
	RadStd		42	75	78	120	143	128	170	95	76	54	38	24

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+2.09	+2.86	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page